

Align Tags — Rule Book

Version 1.0 · 12 August 2026 · Every rule below is field-tested in Revit and enforced by the automated test suite (390 tests). Auto Tag follows the same rules through the shared engine and settings, always at angle 0. To change a rule, quote its number (R1, R2, ...).

1 The Pick

R1 · The pick is the text block's bottom-left corner — on both sides

Wherever you click, the **bottom-left corner of the lowest tag's text** lands there, to the millimetre. The datum never mirrors with the side: left-hand stacks are ragged on their right, right-hand stacks are ragged on their right too, so the flush left column always faces the reader the same way.

Why: one corner, one rule — mirroring the corner with the side made every re-run land differently. The most-litigated rule in the tool; do not change casually.

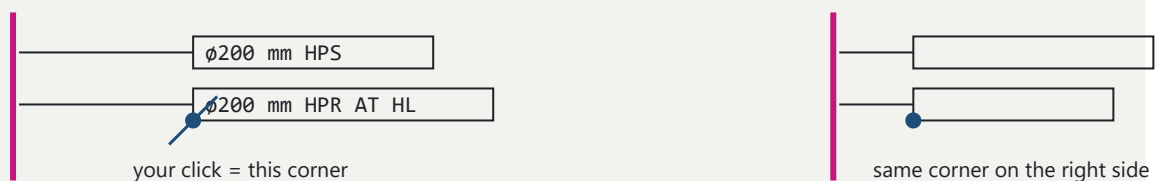


Fig. 1 — the datum corner (dot) is bottom-left on the left side AND on the right side of the pipes.

R2 · The stack always fits — nudged back, never refused

If your pick would put text on the pipes, the stack slides away just far enough: **250 mm clearance** plus the text width. The move is reported (Anchor nudged N mm clear of the pipes). A pick is never rejected.

Why: the old refusal guard rejected 43 picks in one session. A visible, honest nudge beats a dead click.



Fig. 2 — a pick inside the no-fit zone slides back; the distance is printed, never hidden.

R3 · Place, measure, correct — never trust a prediction

The family lies before it draws: boxes include stale leaders, and the text re-anchors with the leader state. So the tool places the stack, **measures the drawn corner**, and re-plans once with the residual — all inside one undo step. Re-running on an already-aligned stack does not shift it.

Why: pre-measured planning landed "internally perfect" stacks hundreds of millimetres off the pick. Drawn geometry is the only truth. Calibrated head-to-edge distances are banked **per project** (0 is a legitimate value); a constant learned in one model is never spent in another.

2 Side and Mode

R4 · Your click decides Left/Right; the dialog decides Upper/Lower

Click left of the tagged pipes' centre → stack left, leaders exit right. Click right → stack right, leaders exit left. The centre reads **each pipe on its own axis** — a riser point contributes its plan position, never its height. The dialog's mode contributes only Upper or Lower.

Why: a height read into the horizontal centre once pinned a riser cluster LEFT for 29 clicks in a row. The side must always be winnable from where you click.

click LEFT → stack left



click RIGHT → stack right



Fig. 3 — one comparison decides the side; Upper/Lower still comes from the saved mode.

R5 · Plain click runs; Shift+click opens the dialog

A plain click on the button aligns with the saved settings. **Shift+click** opens the settings dialog first (the very first run always shows it). Angle 0 selects the straight-leader ruleset (§3); any other angle draws classic slanted leaders. **Auto Tag is pinned to angle 0 permanently** — angled work is a deliberate manual act done with Align Tags.

3 Leader Shapes at Angle 0

R6 · Each tag shapes its leader by its own pipe's direction

Within one stack, every tag looks at the pipe it points to:

Level with a vertical pipe → one straight horizontal leader, elbow grip parked at the line's midpoint.

Reaching a horizontal pipe → horizontal landing, then a true 90° bend onto the pipe. Bends fan out evenly, staying ahead of the text and inside the nearer 60% of the run.

Climbing to a vertical pipe → landing plus a climb leaning 7.5° off vertical, so a leader never draws on top of a pipe.

Arrows keep **250 mm** clear of bends, fittings, pipe ends, and each other; on short pipes the fan redistributes over each pipe's own length, so tags of short pipes sit closer without crowding the long ones.

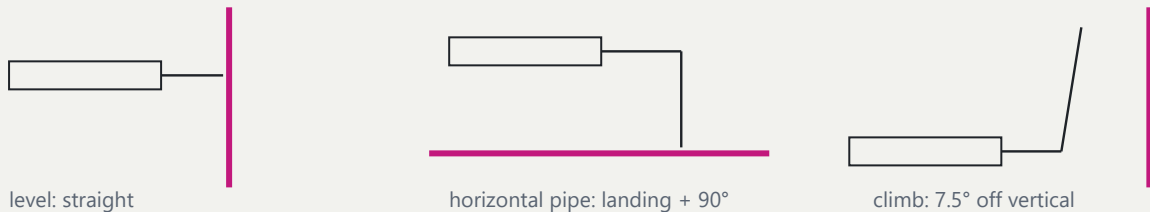


Fig. 4 — the three angle-0 shapes. The 90° bend is exact; the climb's tilt keeps it distinct from pipework.

4 Stack Order

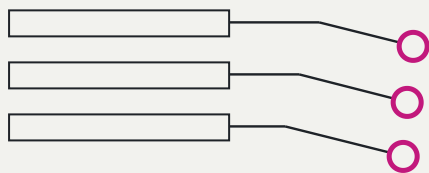
R7 · Runs read like the drawing

Vertical pipes: **leftmost pipe** → **top tag**. Horizontal pipes: **topmost pipe** → **top tag**. The order is the same whichever side the stack sits on.

R8 · Risers tagged from the side: by height, with parallel slants

When the circles sit **beside** the tag rows, the topmost circle takes the top tag. Every leader draws a short landing and then a slant **parallel to the whole cluster's shared angle** — the steepest direct text-to-circle slope, clamped between 15° and 75° — straight into its circle.

Why: ordering by the circles' tiny horizontal stagger produced crossings; per-leader angles read messy. Height order + one shared angle is the approved look.



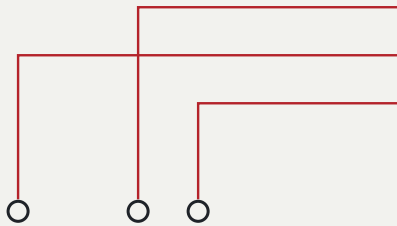
topmost circle ↔ top tag
all slants parallel, one shared angle (15–75°)
bend sits on the landing line

Fig. 5 — side-approach risers: height order, short landings, one parallel sheaf of slants.

R9 · Risers tagged from above or below: by column, nearest to nearest

When the circles sit **wholly above or below** the tag rows, height is noise — the horizontal position decides everything. **The row nearest the pipes takes the drop nearest the stack, marching outward.** For a stack above the circles with leaders exiting left, that is exactly: **leftmost circle → top tag, rightmost → bottom** — and it mirrors itself on the other three sides so no configuration can cross. Each leader lands out to its circle's column and enters with a **clean vertical drop** — no slant.

× height-ordered (circles level: random pairing)



✓ leftmost → top row

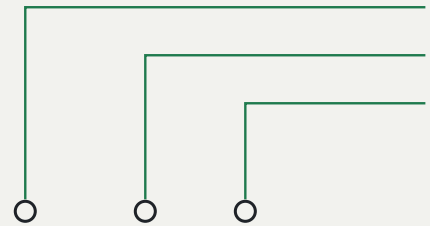


Fig. 6 — drops from above. Left: height order tangles when circles are level. Right: left-to-right down the stack — every landing stops before the drops above it, crossing-free by construction.

5 Clustering — what shares one stack

R10 · A cluster is a physical pipe rack: three tests, all must pass

Two tags share a stack only when their pipes form one rack:

TEST	LIMIT	WHERE SET
1 · Pipes run parallel	within 10°	fixed — unconditional a perpendicular pipe never joins, whatever the other dials say
2 · Pipes sit side-by-side (across the rack)	Rack Width	dialog · default 600 mm
3 · Tagged at the same station (along the run)	Cluster Distance	dialog · 0 = never split

Chaining is transitive — A joins B, B joins C — so a wide group becomes one stack when each *adjacent* gap passes. Clusters cap at **10 tags** (cut at the largest gap); rack bundles have no span cap.

Why the two dials: Cluster Distance works ALONG the run, Rack Width ACROSS it. Runs spaced wider than the rack width tag one by one — raise Rack Width, not Cluster Distance, when parallel pipes refuse to group.

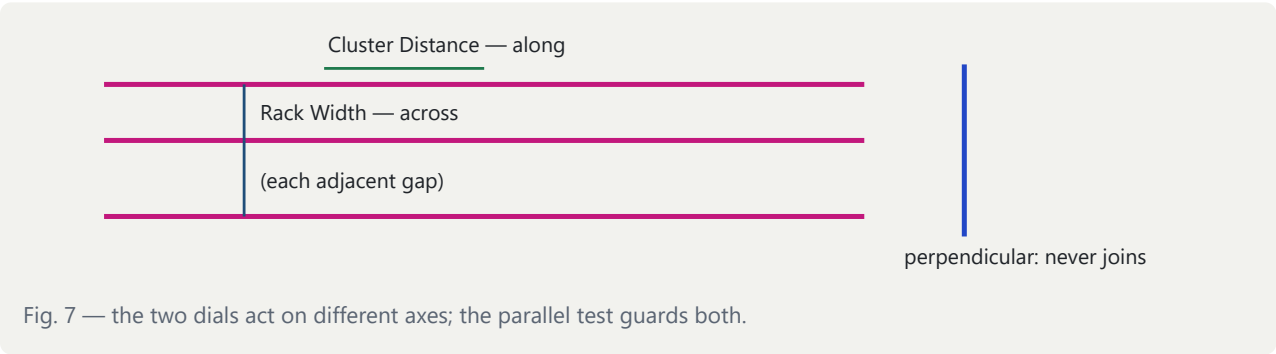


Fig. 7 — the two dials act on different axes; the parallel test guards both.

6 Spacing and Settings

R11 · Row pitch = drawn text height + Vertical Spacing

Vertical Spacing is a **clear gap between texts**, not a centre-to-centre pitch. The tool measures the tallest drawn text in the cluster and adds the gap, so the same setting works at any view scale. All lengths persist in `%APPDATA%\CKR>tag_align_settings.json`, shared by Align Tags and Auto Tag — one dialog tunes both tools.

DIALOG FIELD	MEANING
Vertical Spacing	clear gap between stacked texts
Landing Distance	horizontal run from the text before any bend
Horizontal Spacing	column offset for Intermittent layout only
Cluster Distance	max arrow separation along the run (R10 · test 3); 0 = never split
Rack Width	max pipe separation across the run (R10 · test 2)

7 Exiting the Command

R12 · The exit cleanup may adjust a stack — never override your pick

After the last pick, a cleanup pass resolves overlaps between the stacks placed this session (earlier-placed wins; later yields). Every cluster has a **travel cap of about one text width** from its pick. A conflict that cannot be cleared within the cap sends the cluster **back to your pick**, and the overlap is reported — `N held at their picks (travel cap)` — for manual adjustment. When the later cluster caps out, the earlier one may clear the conflict instead, inside its own cap. One Ctrl+Z reverts the whole cleanup.

Why: the uncapped cleanup once dragged a stack across a garage view on exit, stretching its leaders — the arrows stay pinned on the pipes while the text walks. Your picks are deliberate and land to the millimetre; the cleanup must respect them.

× before: shoved across the view



leaders stretch behind it

✓ now: capped at one text width



max drift ≈ own width

beyond that: snap back + report the conflict

Fig. 8 — the cleanup's leash. Unresolvable overlaps are reported honestly, not "solved" far away.

Changing a rule: quote its number (e.g. "R9 — change the direction for below-stacks"). The rule is changed in code, its tests are updated, this book is edited at `docs/align-tags-rulebook.html`, and the PDF is regenerated — book and behaviour never drift apart.

Sources of authority, in order: this book ← the test suite (tests/) ← the Auto Tag handoff (docs/autotag-align-handoff.md). Where they disagree, the tests reflect what the tool actually does.